TECHNICAL SPECIFICATION

PS-341-360-01-R2

PEP-II RF SYSTEM
2.5 MW KLYSTRON POWER SUPLLY

Prepared by: Paul Bellomo for John DiMaggio — **Date:** 1/27/04
John DiMaggio (Revisor) (Power Conversion Department)

Reviewed by: R Cassel — **Date:** 1/27/04
Richard Cassel (Originator) (Power Conversion Department)

Reviewed by: Joseph Olszewski — **Date:** 1/27/2004
Joseph Olszewski (Power Conversion Department)

Approved by: Paul Bellomo — **Date:** Jan 27, 2004
Paul Bellomo (Power Conversion Department)

File name: 34136001.PS2

PEP2 RF 2.5 MW KLY PWR SPLY REV R2

STANFORD LINEAR ACCELERATOR CENTER
Power Conversion Department
2575 Sand Hill Rd
Menlo park, CA 94025

Technical Specifications

2.5MW PEP-II Klystron
Power supply
Rectifier Transformer
PS-341-360-01-R1

TABLE OF CONTENTS

	Introduction
0.0	Scope, Service and Standards
1.0	Summary of specifications.
2.0	Transformer
3.0	Rectifier Configuration and specifications
4.0	Primary SCR Controls supplied by SLAC
5.0	DC Crowbar supplied by SLAC
7.0	Insulation of complete power supply
8.0	Primary excitation
9.0	Short Circuit Conditions
10.0	Terminals
11.0	Cooling
12.0	Enclosures
13.0	Controls, Accessories and Alarms
14.0	Tests
15.0	Size
16.0	Connection & Interlocks
17.0	Color
18.0	Documentation

Specifications
2.5 MW PEP-II Klystron
Power supply
Rectifier Transformer
PS-341-360-01-R1

Introduction

This specification describes the outdoor oil insulated power supply used to power the PEP-II (B-factory project) 1.2 MW 476 MHz RF SLAC klystron. The power supply shall consist of a phase shifting transformer, rectifier transformers, rectifiers, filter capacitor and inductor and SLAC provided 12 phase primary controller thyristor rectifier, and DC crowbar (powered from the SLAC 12.5 kV 3 phase 60 Hz power line) Figure 1. All of the transformers and high voltage parts of the power supply shall be totally enclosed in outdoor oil tank which is sized to fit the **existing PEP transformer pads** (Figure 9).

0.0 Scope, Service and Standards

0.1 The scope encompasses the **design, fabrication, test and delivery** of the power supply system in accordance with this specification, including designing, performing thermal, stress and seismic analyses, harmonic and short circuit current calculations to demonstrate compliance with specification, and providing installation and maintenance manuals and sufficient drawings for the power supply and transformers operation and maintenance.

0.2 The vendor shall be responsible for the testing of all installed equipment and performance of all equipment supplied by the vendor.

0.3 The following equipment **as a minimum** shall be supplied by vendor.

1) One transformer tank with monitoring and controls enclosure.

2) One Phase shifting transformer with input current transformers.

3) Two Rectifier transformers.

4) Two Filter inductors.

5) Five output Bushings.

0.4 In addition the following equipment shall be installed by the vendor.

1) Installation of the SLAC supplied Primary Control SCRs.

2) Installation of the SLAC supplied SCR crowbar.

3) Installation of the four sets of rectifiers, filter capacitors and interconnecting resistors.

0.5 The following equipment may be supplied by the vendor or can be supplied by SLAC.

1) Four sets of 6 pulse high voltage power rectifiers.

2) Four sets of 6 pulse high voltage filter rectifiers.

3) Four filter capacitors with interconnecting resistors.

0.6 The equipment furnished under this specification shall comply with the currently applicable accepted industry practice, in particular with the following:

IEEE Std General Principles for Temperature Limits in the Rating of Electric Equipment.

ANSI C42.100 1972 Dictionary of Electrical and Electronic Terms.

AEC Safety & Fire Protection Bulletin #13 Electrical Safety Guides for Research, Part II.

Title 29, Chapter XVII, Part 1910 - Occupational and Health Standards.

National Electrical Code National Fire Protection Association Number 70

National Electrical Safety Code, ANSI C2.

ANSI C34.2 Practice and Requirements for Semi-Conductor Power Rectifiers.

IEEE Std 4-1969 Techniques for Dielectric Tests.

ANSI/IEEE C57.98-1986	Transformer impulse testing
ANSI C57.12	Power Transformers.
ANSI C57.18	Rectifier Transformers.

Uniform Building Code California, Pacific Coast Building Conference.

1.0 **Summary of specifications**

1.1 There shall be five outputs for a possible future depressed collector klystron:

a) Cathode #5	0 to -90 kV
c) Output #4	0 to -77 kV (86% of cathode voltage)
d) Output #3	0 to -52 kV (57% of cathode voltage)
e) Output #2	0 to -26 kV (29% of cathode voltage)
b) Collector #1 Nominal Ground	0 to -1 kV maximum

1.2 The maximum output current on **any** output is 27 Amps. (Note: the klystron cathode beam current is 27 amps maximum and the current may return in any of the outputs or any combination of outputs)

1.3	Frequency line	60 HZ
1.4	Line voltage	12.5 kilovolts ±5% line to line
1.5	**Nominal** phase shifting transformer power	350 kVA (see figure 2)
1.6	Rectifier transformer primary voltage	12.5 +5% kilovolts line to line
1.7	Each secondary voltage	21 kV RMS. line to line or 10.5 kV (T2-S2) See figure 3
1.8	Each Secondary Current	22 Amperes RMS. per phase
1.9	**Nominal** rectifier transformer power rating	1500 kVA (each of two transformers)
1.10	Filter choke inductance	>0.3 Hy at 85 amps DC
1.11	Filter choke DC current, long term	85 Amps

2.0 **Transformer**

2.1 The phase shifting transformer shall be connected to provide for a phase shift line to primary of ±15° (±0.5°) (See figure 2).

2.2 The phase shifting transformer shall have a minimum of two monitor windings (for SCR synchronization) connected to the terminal box. The voltage ratio shall be approximately 100/1, i.e. 125 volts line to line for 12.5 kv line to line. The current rating shall be 5 amps RMS.

2.3 The rectifier transformers shall be Wye secondary with open Wye primary.

2.4 The secondary of the transformers shall have an additional +5% voltage tap used by the proprietary filter circuit. (See figure 3)

2.5 The secondary voltage winding shall be 21 kV RMS. line to line for T1-S1, T1-S2 , T2-S1. and 10.5 kV RMS. line to line for T2-S2 with a primary voltage of 12.5 kv RMS. line to line.

2.6 The transformers secondary (they can be different) shall have a nominal leakage reactance to the primary of between 7% and 10% based on the **nominal kVA rating** of the transformer.

2.7 The leakage reactance from phase winding to phase winding may vary within ±7.5% of the nominal leakage reactance.

2.8 The resistive impedance shall be < 2%, based on the **nominal kVA rating**.

3.0 Rectifier Configuration and specifications

The vendor may propose his own rectifier, filter capacitor and resistor design and fabrication, as per following specification, or **may install SLAC supplied rectifiers**.

3.1 The supply shall have 8 sets of 6 pulse full wave rectifiers.

3.2 The 4 sets of power 6 pulse full wave rectifiers shall be rated for 30 Amps average with a junction temperature not to exceed 80° C with a 45° C ambient air temperature at full load.

3.3 The individual rectifier stacks shall have a continuous reverse voltage rating of not less than 30 kV and be balanced / protected so as not to exceed the individual diode's peak inverse voltage rating with 500 Amps in the reverse direction through the protection.

3.4 The diodes shall be rated for a fault current consistent with a bolted secondary or (crowbar) with SCRs phased full on and an 8 cycle breaker interruption. The junction temperature shall not exceed 150° C or the manufacturer's maximum operating junction temperature rating, under full load operating conditions with a 45° C ambient air temperature.

3.5 The 4 sets of filter 6 pulse full wave rectifier shall have diodes rated at >3 amp average used for ripple filtering. The junction temperature shall not exceed 80° C with a 45° C ambient air temperature at full load and at a fault current of 25 amps for 8 cycles.

3.6 The filter rectifier stacks shall have a continuous voltage rating of not less than 30 kV and shall be balanced or protected so as not to exceed the individual diode's peak inverse voltage rating with less than 50 kV per stack.

3.7 The supply shall have 4 filter capacitors of not less than 8 uF at 30 kV DC each.

3.8 The filter shall also consist of 8 resistors or resistor strings rated at 500 ohms and 1 kW in oil with energy absorption capacity of no less than 5000 Joules per resistor.

3.9 The capacitance between each secondary and shield or primary shall not exceed 0.005 uF.

3.10 The power loss, if SLAC provides the rectifiers, filter and filter resistors shall not exceed 10 kw

4.0 **Primary SCR Controls supplied by SLAC**

The vendor shall install, make all connection to, and test the SLAC supplied 12 phase SCR's snubbers, SCR drivers and control system. (see Figure 6). Detail drawings with dimensions will be available for the vendor two weeks before the required vendors design review. The pre-tested parts shall be available from SLAC at least 4 weeks before the required power supply delivery.

4.1 The supply shall be 12 phase primary controlled SCR in an open Wye configuration.

4.2 The SCR enclosure shall have its own separate oil system. The enclosure will be vented into the main oil tank and shall be by conduction / convection cooled from the main tank oil system and/or pump oil provided by SLAC. The maximum power loss in the SCR enclosure shall be 6000 watts.

4.3 Acces to the primary SCRs for maintenance or repair shall be through a removable cover on the top supplied by SLAC. The transformer shall be design in such a way so as not to impede the access to the SCRs and minimize interference to allow for removal of the whole phase SCR system as a unit.

5.0 DC Crowbar supplied by SLAC

The vendor shall install, make all connections to and test the SLAC supplied SCR crowbar, Cable snubbers, voltage dividers, SCR drivers and control system (see Figure 7). Detail drawings with dimensions will be available for the vendor two weeks before the required vendors design review. The pre-tested parts shall be available from SLAC at least 4 weeks before the required power supply delivery.

5.1 The supply shall have 4 SCR type crowbars (one for each of the power supply section).

5.2 Each crowbar shall be capable of shorting the supply in less than 5 μseconds from an external command or if the klystron arcs down.

5.3 The SCR crowbar enclosure shall have its own separate oil system. The enclosure will be vented into the main oil tank and shall be by conduction / convection cooled from the main tank oil system. The maximum power loss in the SCR crowbar enclosure shall be 500 watts.

5.4 Access to the crowbar SCRs for maintenance or repair shall be through a removable cover on the top supplied by SLAC. The transformer shall be designed in such a way so as not to impede the access to the SCRs and minimize interference to allow for removal of the whole crowbar SCR system as a unit.

6.0 **Filter inductors**

6.1 There shall be two filter inductors operating at 360 Hz, 6 pulse rectification.

6.2 The inductor shall be a so called "swinging choke" or saturable choke with incremental inductance versus current between the maximum and minimum values shown in figure 8.

6.3 Each of the two filter inductors shall have a filter inductor voltage monitoring winding connected to the terminal box. The turns ratio shall be approximately 100/1 or 120 V for 12 kV across the inductor. The current rating shall be 5 amps RMS.

6.4 The inductance of each filter inductor shall be >0.3 Hy at 85 amps DC.

6.5	DC current long, term	85 amps
6.6	Fundamental ripple frequency	360 HZ
6.7	Ripple voltage RMS.	6.5 kV max., normal 1.5 kV
6.8	Maximum voltage	12.5 kV RMS. at 60 Hz
6.9	Maximum fault current	1000 amps 135 mSec.
6.10	Inductance at 200 amps	>0.02 Hy
6.11	High potential test	20 kV RMS., 60 Hz for one minute
6.12	Audible noise level at 20 feet	<55 dbA

7.0 **Insulation of complete power supply**

7.1 The insulation for the primary windings, current transformers, and terminal bushings shall be 15 kV class 95 kV BIL (underground distribution, grounded neutral).

7.2 The phase shifting transformer and rectifier transformers primary and the filter inductors shall withstand a 60 Hz high pot test of 34 kV for one minute with SCR phased control on.

7.3 One secondary of the rectifier transformers will operate with a DC voltage of 35 kV superimposed on the 21 kV AC voltage and therefore shall have a **DC high pot** of 80 kV DC **and/or** 60 kV DC with rectifier in place.

7.4 The other rectifier transformer secondary shall have an 85 kV DC voltage superimposed on the 10-21 kV AC voltage and therefore shall have a **DC high pot** of 150 kV DC **and/or** 120 kV with rectifier in place.

8.0 Primary excitation

8.1 The phase shifting transformer shall be designed for continuous operation with 12.5 kV +5% RMS. at 60 Hz.

8.2 The SCR's in the open Wye will control the primary excitation voltage on the rectifier transformers so **as not to exceed the equivalent of a 13 kV 60 Hz line to line excitation** under all conditions. The vendor shall designed the rectifier transformers for an excitation (Volt-seconds) equivalent to **13 kV RMS. 60 Hz Line to line. (12.5 kV/21 kV for turns ratio)** {**NOTE** DC in primary winding will be limited to 0.5 Amp}. The total core losses (eddy currents and hysteresis) under SCR primary control shall be assumed by the vendor to be not more 10% greater than the total core losses experienced under 13 kV RMS. 60 Hz excitation (without the SCR primary control, non-grounded wye, no zero sequence current).

9.0 Short Circuit Conditions

9.1 The transformer shall be designed for continuous full load operation.

9.2 The transformers shall withstand an external 3 phase secondary short circuits at the transformer terminals for 5 cycles without appreciable damage or parameter change.

9.3 The transformers shall be designed for frequent crowbars (secondary short circuits) with interruption time from the primary SCR's of less than one cycle. The number of crowbars will not exceed 10,000 with no more than (10) in any one day.

10.0 **Terminals**

10.1 The primary line terminal shall be made compatible with SLAC's existing disconnect and vacuum contactor enclosure utilizing a weatherproof throat connection. Final dimensions and placement to be determined before design review.

10.2 The open Wye terminals, filter inductor terminals, and SCR crowbar terminals (supplied by SLAC) are to be connected by the vendor. The terminals shall be leak tight oil to oil feed through. The details will be made available before the vendors required design review.

10.3 The rectifier transformer secondary terminals, consisting of the 3 phases of each Wye and the three +5% taps on each Wye, shall be terminated to the rectifier diodes on insulators in such a way that tools or debris will not fall into the transformers if dropped when working on the rectifiers.

10.4 The five DC Output terminals shall be sealed oil tight with cable termination for RG-220 Cable. The low voltage terminals shall be Isolation Design Inc. Model D-130-02 and the High voltage terminal Model D-117-BA or SLAC approved equal.

11.0 **Cooling and oil**

11.1 The power supply shall be forced oil /convection air cooling (FO) using an oil pump to insure adequate cooling for the transformers and the rectifiers and SCR at the top of the tank.

11.2 The oil pump(s) shall be designed to allow for oil flow in the air cooled radiators without the pump(s) being on. Pump(s) shall be arrange to enable removal and maintenance on the pumps without draining the oil from the tank.

11.3 The average winding temperature rise of all transformers and filter inductors shall be 65°C maximum with a hot spot temperature rise of 80°C maximum above 40°C ambient.

11.4 The transformers and inductors shall comply with the loading guide for 65° oil transformers NEMA standards.

11.5 The oil used in the transformer shall be Sontex® X-ray oil {Drakeol® 10 LT mineral Oil NF} or approved equal.

12.0 **Enclosures**

12.1 The transformers enclosure shall be of the vendor's standard design. It shall however meet industrial standards for transformer tank design. It shall have sufficient space for all of the vendor supplied equipment as well as for the SCR phase controllers and crowbar enclosures supplied by SLAC.

12.2 The transformer enclosure shall be capable of withstanding a full vacuum for oil filling and processing. The SCR enclosures will be vented into the main tank for oil overflow and vacuum oil filling.

12.3 The diode rectifier section shall be located on the top (figure 4) of the transformer and shall be a minimum of 48" W x 82" L x 24" H. This will contain the rectifier diodes, filter capacitors, and filter resistors. The rectifier section shall be separated from the rectifier transformers so that it can be installation and maintenance without draining the tank oil and designed in such a way that tools or debris will not lodge in the transformers if dropped from the diode or filter.

12.4 The space shall be provided for the SLAC supplied SCR controller shall be a minimum of 54"W x 36"H x 19"D inside the tank with a flange of at least 2" wide at the top (see figure 6). The SLAC supplied SCR controller will come as a unit and shall be installed in such a way that it can be removed easily and quickly as a unit without draining the main tank oil.

12.5 The space shall be provided for the SLAC supplied SCR crowbar shall be a minimum of 54"W x 36"H x 19"D inside the tank with a flange of at least 2" wide at the top (see figure 7).

The SLAC supplied SCR crowbar will come as a unit and shall be installed in such a way that it can be removed easily and quickly as a unit without draining the main tank oil.

12.6 The transformer enclosure shall be designed to have a removable "cat walk" used for personnel safety during installation and subsequent maintenance of the rectifiers and SCR units.

13.0 Controls, Accessories and Alarms

13.1 The voltage control shall be provided by the SCR primary control. All of the system controls except those listed in 13.4 will be provide by SLAC.

13.2 The crowbar shall be triggered from the SLAC provided control system.

13.3 Each of the primary bushings, shall have current transformers type 300/5 with ASA accuracy classes 1.2 and be wired to a terminal box on the outside of the transformer.

13.4 The following alarm devices shall be furnished and wired to a weatherproof terminal box. Each device shall have at least one NO and one NC contact.

A) Pressure relief devices with two independent sets of contacts, Qualitol Series 208 or SLAC approved equal.

B) Top oil temperature for main tank Qualitol Series 165 or SLAC approved equal.

C) Oil level gage for main tank Qualitol Series 032-35 or SLAC approved equal.

D) Pressure / Vacuum alarm.

E) Oil flow Indicator.

14.0 Tests

14.1 The following factory tests shall be performed on the power supply or transformers in accordance with IEEE and ANSI standards.

14.2 Resistance measurement.

14.3 Transformer turns ratio tests.

14.4 Polarity & phase relationship.

14.5 Impedance.

14.6 Induced potential tests.

14.7 AC high pot test on the primary.

14.8 DC high pot test on the secondary and rectifier system.

14.9 Temperature rise tests. (one unit)

14.10 BIL test 95 kV full wave 1½-40 usec on the primaries ANSI/IEEE C57.98-1986 (first unit)

14.11 Incremental inductance vs. current for filter inductors at 360 hz

15.0 Size

15.1 The overall size of the transformer shall not exceed 144" wide x 94" deep x 144" high, with pad area not to exceed 130" wide x 84" deep (See Figures 4 & 9).

15.2 The detail size and design of the SCR and Crowbar enclosure will be mutually agreed upon before the required vendor design review.

15.3. Details dimension of the 12 kV throat and the DC terminations will be mutually agreed upon during the design stage.

15.4. Provision shall be made to lift and handling the transformer complete with oil by a crane.

15.5. Anchoring for earthquake protection shall also be provided.

16.0 Connection & Interlocks

This section covers the interlocks and connection required.

16.1 All of the power connections shall be totally enclosed in metal and secured by either welding or at least 4 bolts, 1/4" or greater adequate for outdoor service and for personal safety.

16.2 All 15 kV class terminals and connections shall conform with the ASA and NEMA standards where they apply.

17.0 **Color**

17.1 The transformers shall have a final color of "Home Spun Brown" as supplied by Tnemec Company of Napa, California color # l6794 or SLAC approved equivalent.

18.0 **Documentation**

The vendor shall submit the below listed documentation with each Power Supply in an electronic file format (.dxf, .dwg or .dgn).

18.1 All assembly and sub-assembly drawings and schematics.

18.2 A complete parts list with manufacturer's name and model/part number.

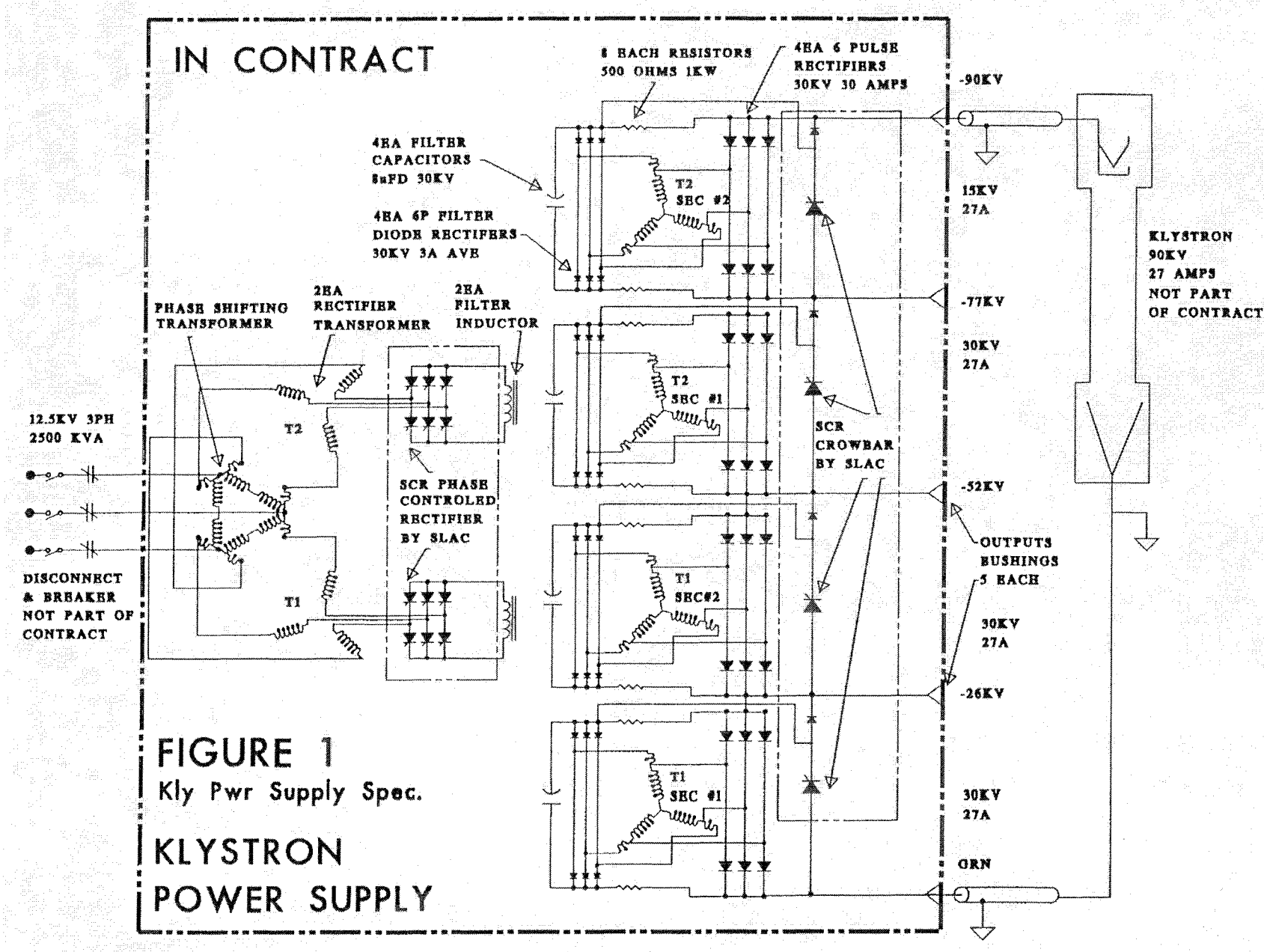
18.3 Test procedures and test results.

18.4 Installation and Maintenance Manuals.

18.5 This requirement is deleted.

18.6 This requirement is deleted.

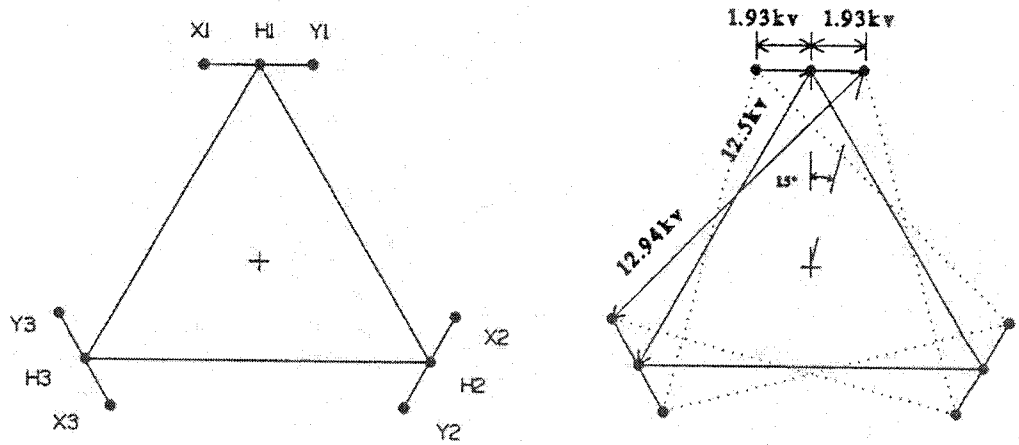
18.7 Vendors in house technical specifications.


IN CONTRACT
8 EACH RESISTORS
500 OHMS 1KW
4EA 6 PULSE
RECTIFIERS
30KV 30 AMPS
-90KV
4EA FILTER
CAPACITORS
8uFD 30KV
4EA 6P FILTER
DIODE RECTIFIERS
30KV 3A AVE
T2
SEC #2
15KV
27A
KLYSTRON
90KV
27 AMPS
NOT PART
OF CONTRACT
PHASE SHIFTING
TRANSFORMER
2EA
RECTIFIER
TRANSFORMER
2EA
FILTER
INDUCTOR
-77KV
30KV
27A
T2
SEC #1
T2
12.5KV 3PH
2500 KVA
SCR
CROWBAR
BY SLAC
SCR PHASE
CONTROLED
RECTIFIER
BY SLAC
-52KV
OUTPUTS
BUSHINGS
5 EACH
T1
SEC#2
T1
DISCONNECT
& BREAKER
NOT PART OF
CONTRACT
30KV
27A
-26KV
T1
SEC #1
30KV
27A
GRN
FIGURE 1
Kly Pwr Supply Spec.
KLYSTRON
POWER SUPPLY

PHASE SHIFTING TRANSFORMER

PRIMARY WINDING CURRENT (H1-H2) < 10 AMPS RMS

INPUT (H) 127 AMPS RMS, OUTPUTS (X & Y) < 71 AMPS RMS



+- 15 DEGREES 12.5KV 3 PHASE

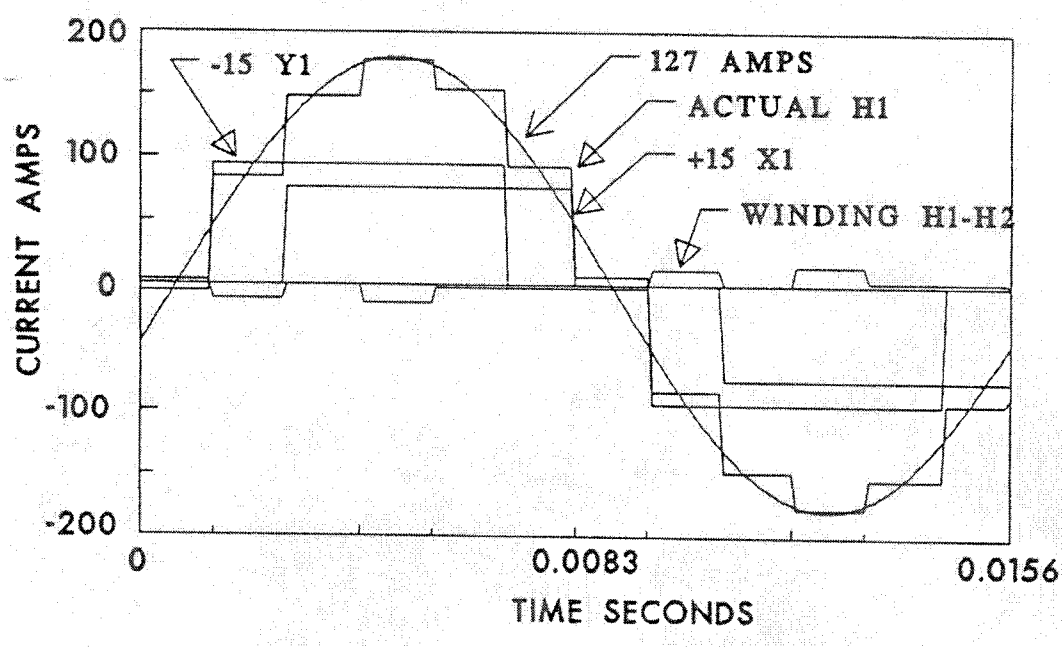
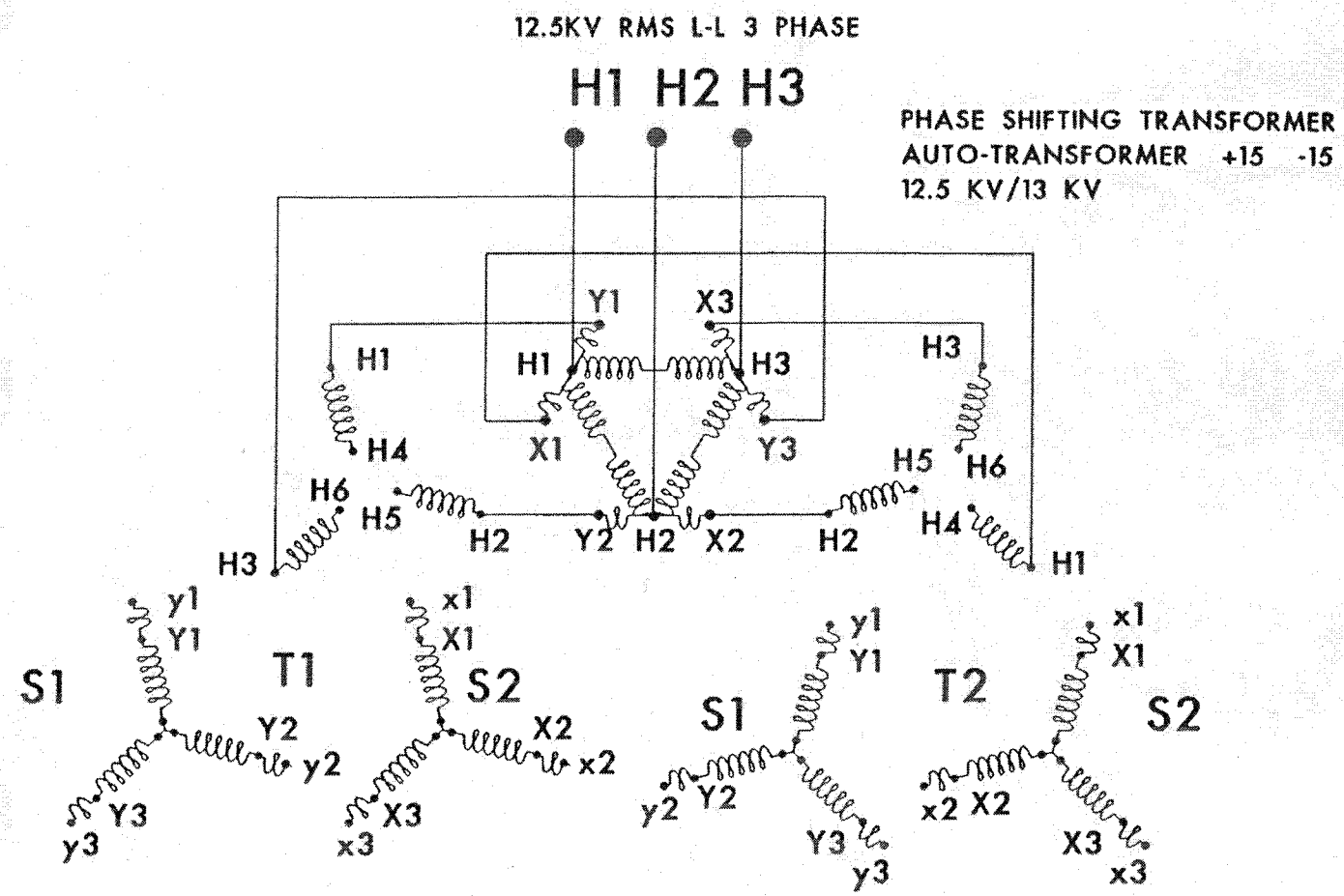


FIGURE 2

Kly Pwr Supply Spec.

TRANSFORMR CONNECTION DIAGRAM



RECTIFIER TRANSFORMER

T1 S1 750 KVA 21KV 22AMPS 3 PHASE
T1 S2 750 KVA 21KV 22AMPS 3 PHASE
T2 S1 750 KVA 21KV 22AMPS 3 PHASE
T2 S2 375 KVA 10.5KV 22AMPS 3 PHASE

Figure 3

Kly Pwr Supply spec.

PEP-II 2.5 MW KLYSTRON POWER SUPPLY LAYOUT

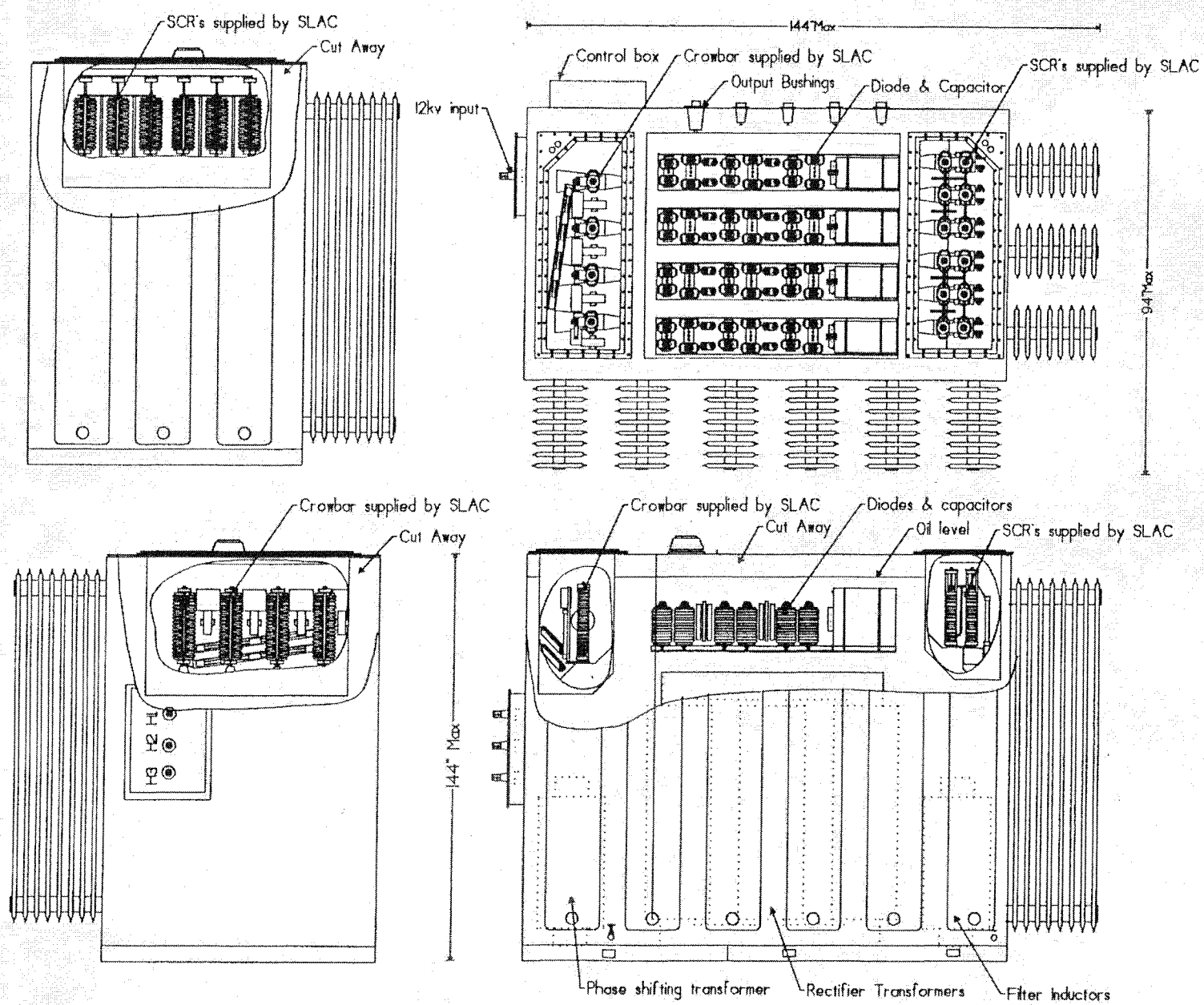


FIGURE 4

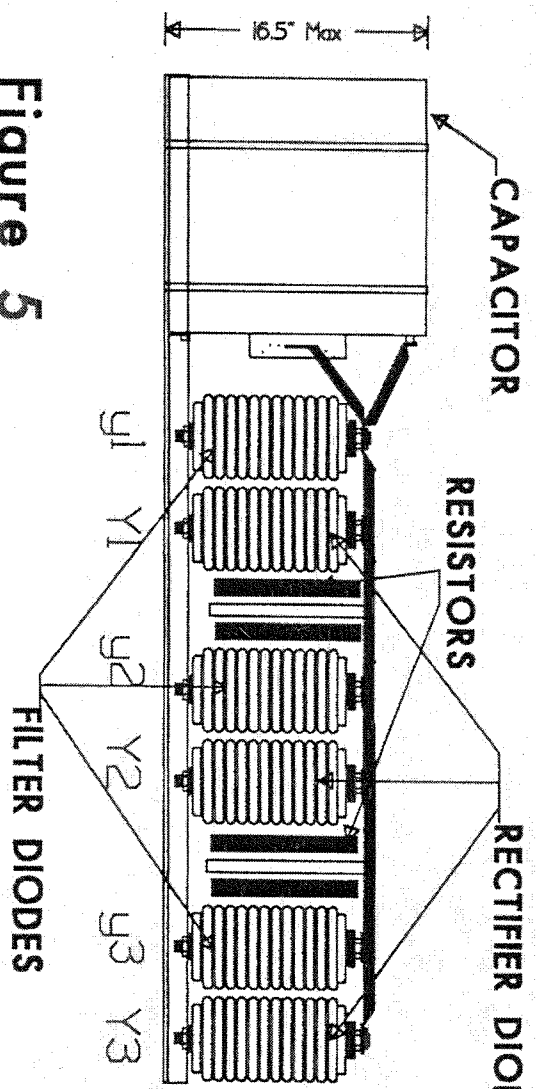


Figure 5
Kly Pwr Supply Spec.

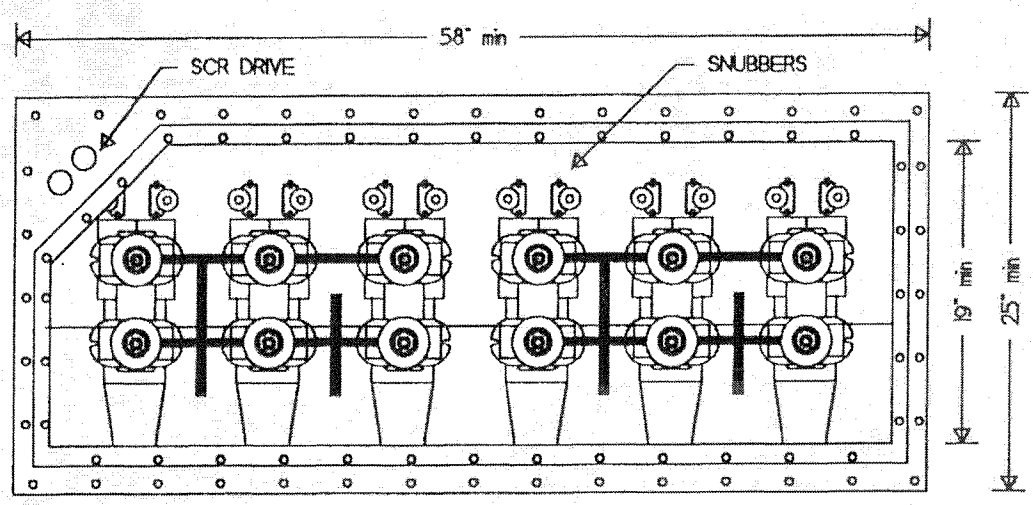
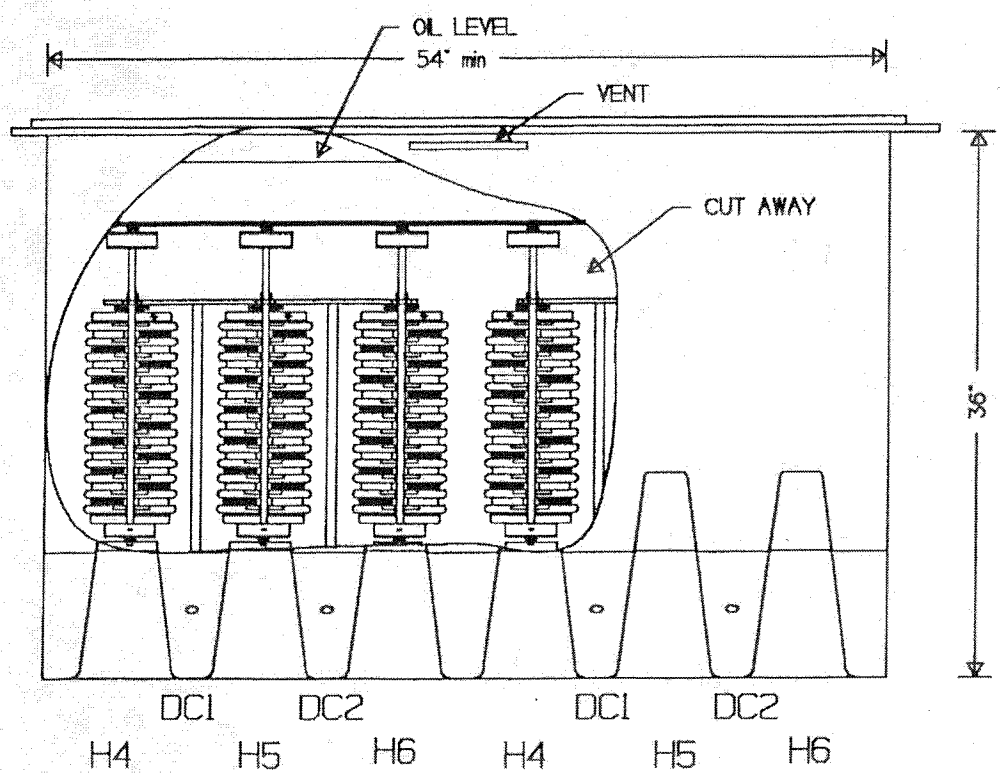
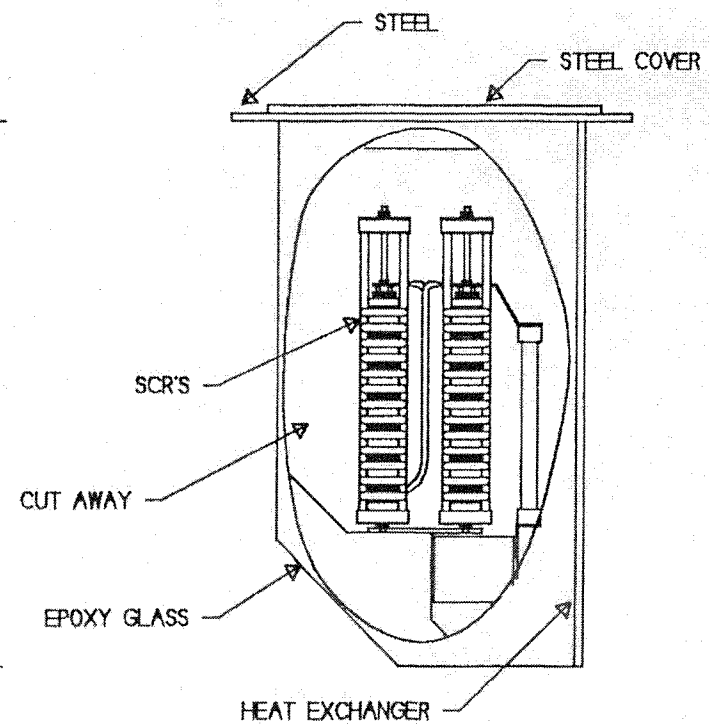
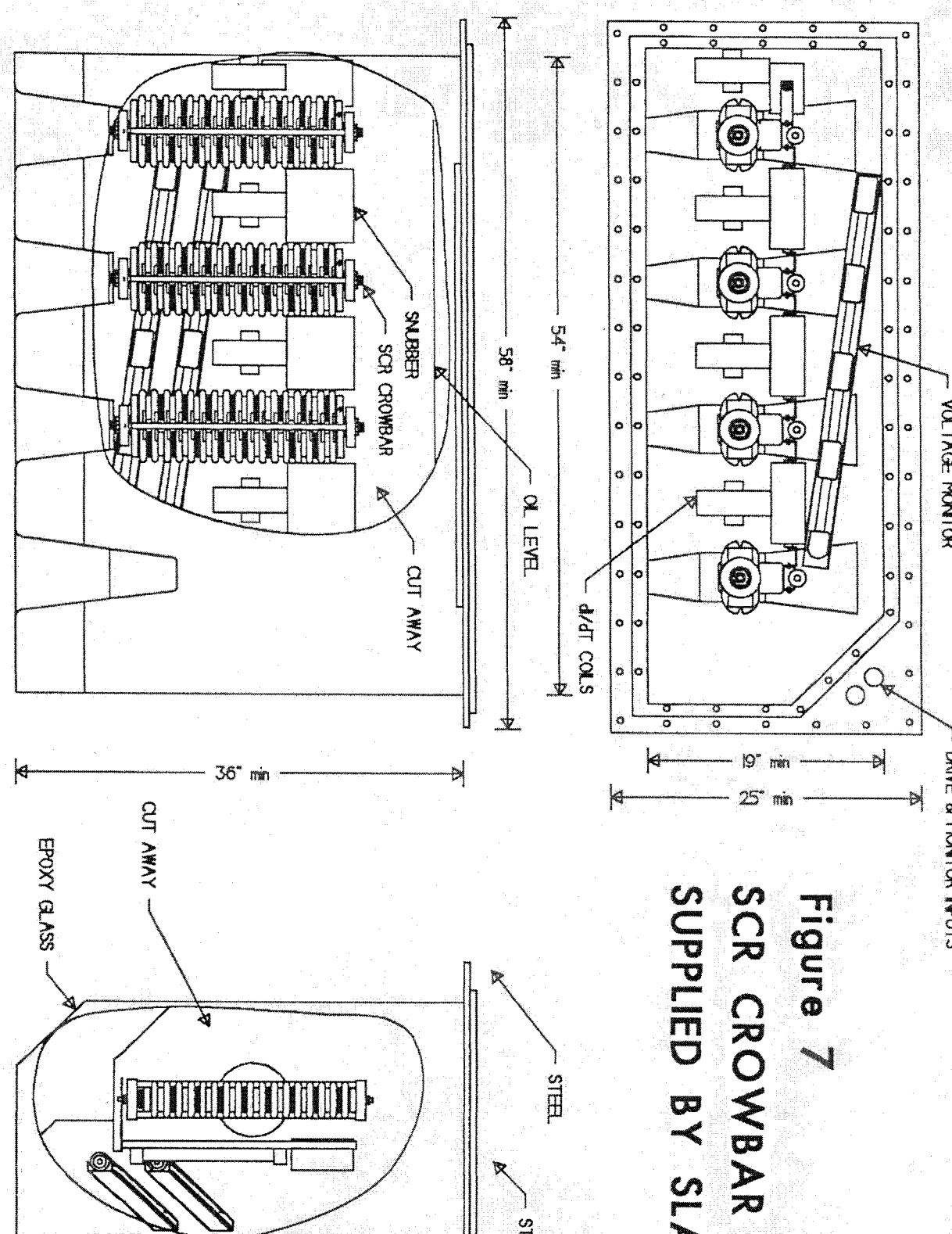


Figure 6
12kv SCR controller supplied by SLAC







26KV 52KV 78KV 90 KV

Figure 7
SCR CROWBAR
SUPPLIED BY SLAC

FILTER INDUCTOR

INDUCTANCE VS CURRENT

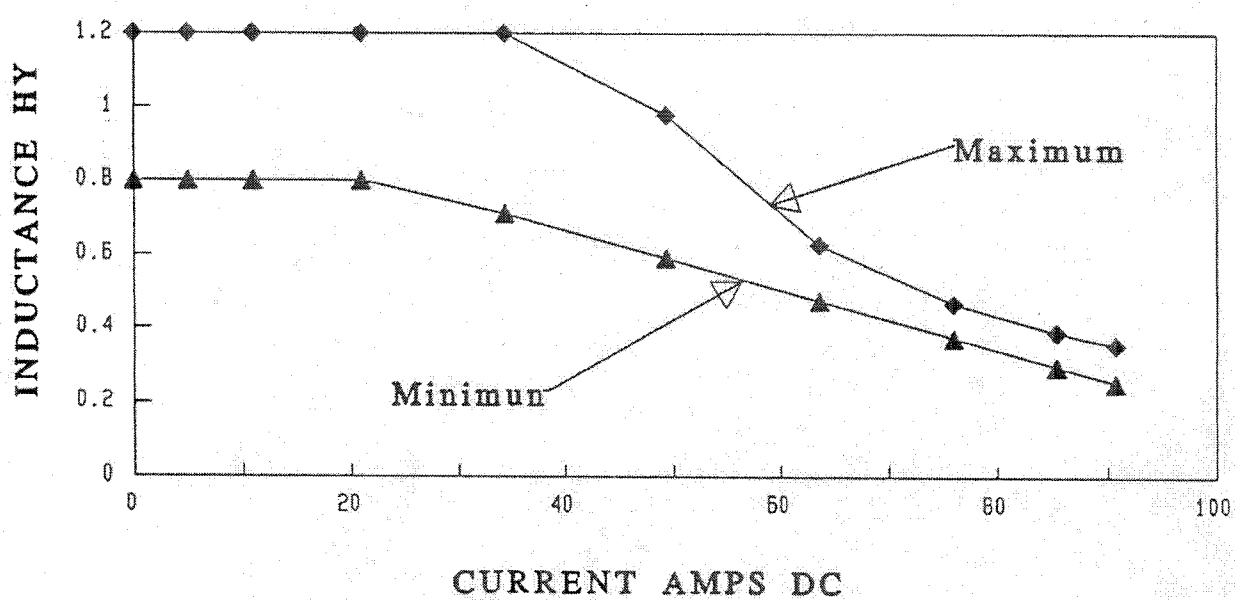


Figure 8

PEP-II KLYSTRON POWER SUPPLY LAYOUT

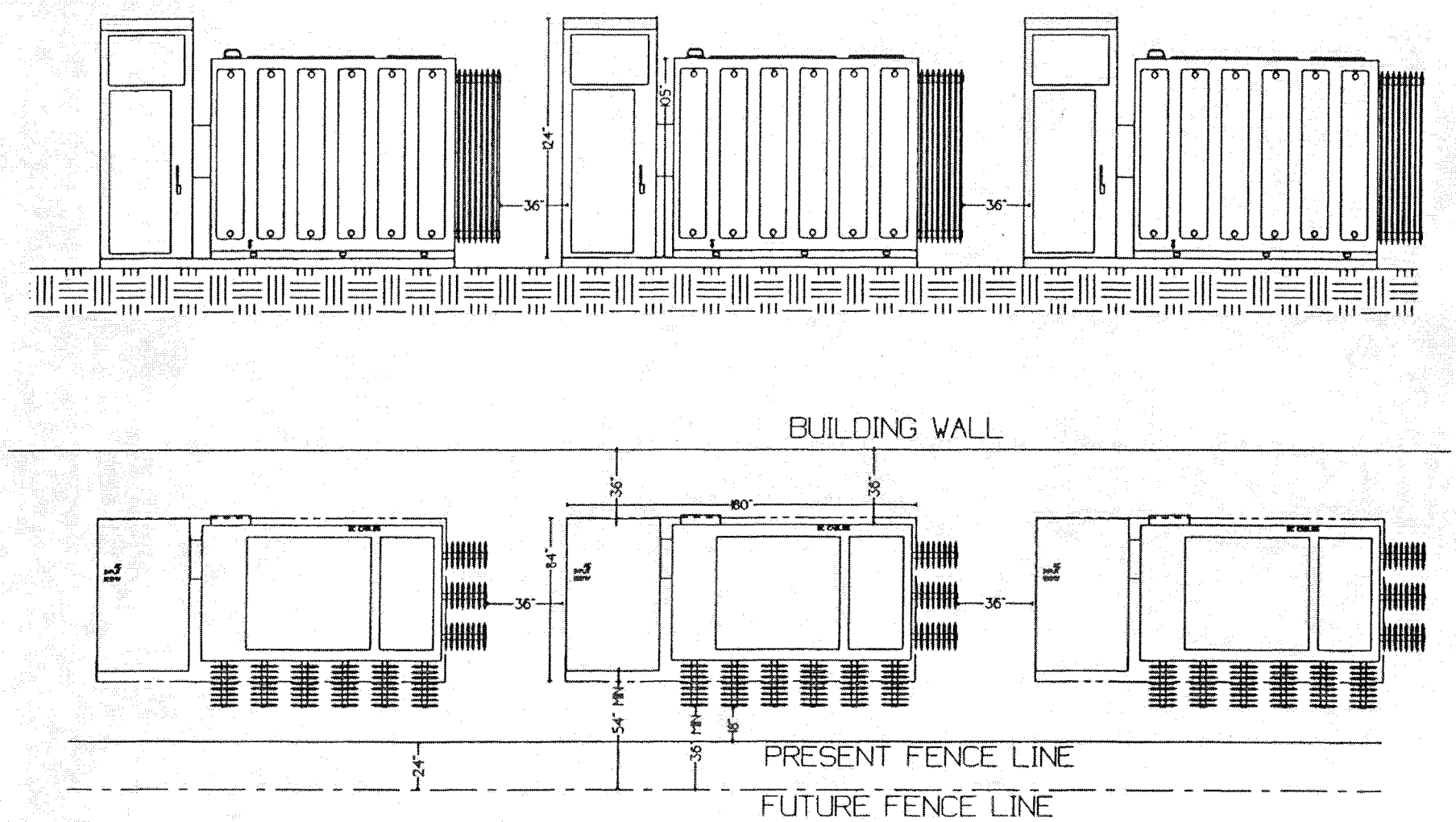


Figure 9